

Setu SCX Single-Channel Indoor LoRa® Gateway Datasheet

Product Overview

Setu SCX is a cost-efficient and compact **single-channel Indoor LoRa® gateway** designed for small-scale IoT deployments inside commercial and enterprise environments. Engineered for seamless and reliable wireless coverage in localized spaces, it acts as the perfect network extender for smart buildings and industrial interiors.

With **Wi-Fi and Cellular backhaul connectivity**, Setu SCX eliminates the need for wired infrastructure—making it ideal for installations in places where Ethernet cabling is not available or practical. The ultra-low-power, fanless design ensures silent operation in offices, retail stores, hospitals, hotels, and indoor utility rooms.

Setu SCX supports **AES-128 encrypted communication**, OTA configuration and monitoring through MQTT / HTTPS secure and scalable network performance.

Whether enabling indoor environmental monitoring, asset tracking, energy management, occupancy sensing, or small-area automation—**Setu SCX provides a simple, powerful, and plug-and-play LoRa® network entry point** for rapid IoT adoption.





Key Features

Compact Single-Channel LoRa® Gateway

Designed for localized indoor IoT deployments with small device density and focused wireless coverage.

Plug-and-Play Installation

Simple onboarding using MacCloud, DHCP auto-connect and rapid provisioning for non-technical deployment teams.

Fanless Silent Operation

Zero-noise design ideal for commercial indoor spaces like offices, hospitals and retail stores.

Universal Server Compatibility

Works with private network servers with MQTT, HTTP, HTTPS integration.

Dual Backhaul Connectivity

Wi-Fi and Cellular connectivity for environments without Ethernet cabling or wired network access.

Enterprise-Grade Security

AES-128 encrypted LoRa® communication with secure TLS-based OTA configuration and device authentication.

Low-Power Architecture

Efficient operation optimized for 24/7 uptime while reducing overall infrastructure cost.

Flexible Mounting

Wall and ceiling installation supported for optimized RF propagation inside buildings.

Suited Applications

Smart Buildings & Offices

Indoor LoRa® connectivity hub for BMS automation, energy management and workplace monitoring.

Retail & Shopping Centers

Supports people counting, occupancy analytics and warehouse-to-store supply chain tracking.

Hospitals & Healthcare Facilities

Reliable gateway for indoor asset tracking, patient safety systems and environmental compliance.

Hotels & Hospitality Spaces

Enables smart room controls, environmental sensors and maintenance alerts for enhanced guest experience.

Smart Campuses & Education

Wide indoor-area gateway for security, environment sensing and resource optimization in institutions.

Data Centers & Server Rooms

Ideal for temperature, humidity and access monitoring in controlled indoor areas.

Public Venues & Indoor Infrastructure

Reliable LoRa® coverage for airports, metros and community facilities.

Industry 4.0 & Factory Floors

Connects predictive maintenance sensors and equipment monitoring systems across industrial interiors.

Warehouses & Inventory Areas

Backbone for indoor logistics, RFID extensions and smart storage infrastructure.

Technical Specifications

General Specifications

Part Order Number	Setu SCX
Product Brand	Macnman Technologies Pvt Ltd

System Hardware Specifications

Processor Type	nRF5340 Dual-Core Bluetooth® LE & LoRa Gateway MCU
Number of Cores	Dual-Core (1× Application + 1× Network Core)
Clock Speed	128 MHz (Application Core) / 64 MHz (Network Core)
Memory Size	512 KB RAM
Memory Architecture	Low-Power Multi-Core Shared Memory Architecture
Internal Flash	1 MB On-Chip Flash + External Flash Support
USB Port	Debug Interface (Bottom of Enclosure)
Connector Type	USB Type-C
Interface Protection	Surge Protection Enabled

LoRa Specifications

Compatibility	Macnman LoRa Nodes & Controllers
Supported Frequency Bands	IN865

Channel Support	Single-Channel (1 × Uplink / 1 × Downlink)
Data-Rate	Fixed Data-Rate – No ADR (single-channel limitation)
Tx Power	+23 dBm @ IN865/868 MHz +20 dBm @ 915 MHz
Rx Sensitivity	Down to –136 dBm @ SF12, BW 125 kHz
Antenna Type	External High-Efficiency LoRa Antenna
Device Capacity	Optimized for Localized Indoor Installations with Small Node Density

Backhual WiFi

Bands	2.4 GHz ISM Band 5 GHz ISM Band
Standards	IEEE 802.11 b/g/n, 802.3, 802.3u, 802.11e-compatible bursting
Modulation Schemes	BPSK, QPSK, 16 QAM, 64 QAM
Security	WEP, TKIP, AES, WPA, WPA2
Antenna	External Antenna Integrated in Enclosure

Backhual Cellular Lte*

Cellular Bands	B1 / B3 / B5 / B7 / B8 / B20 / B28
Standards	3GPP LTE, IEEE 802.3, IEEE 802.3u, 802.11e-compatible bursting
Modulation Schemes	BPSK, QPSK, 16-QAM, 64-QAM
Security	WEP, TKIP, AES, WPA, WPA2
Antenna	Internal Antenna Integrated in Enclosure

Supported Protocols

MQTT	HTTP HTTPS
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Physical Characteristics

Power Supply	9-36 V PoE
Operating Temperature	-40°C to +80°C
Humidity Range	0% to 100% RH (non-condensing)

Ingress Protection	IP35
Housing Material	Aluminium
Color	Matte White
Weight	~190g
Mounting Options(Brackets)	Pole Mounting / Wall Mounting

Certifications & Compliance

WPC	Compliant
RoHS	Conforms to RoHS directive

Security & Encryption

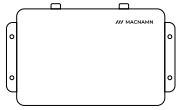
Data Encryption	AES-128 end-to-end encryption
Authentication	OTAA activation modes (Nodes)
Firmware Protection	Secure boot and signed OTA updates

Setu SCX Variants

VARIANT NAME	POWER SOURCE	WIFI	ETHERNET	CELLULAR	IDEAL FOR
Setu SCX1	12–24 V (Power)				Ideal for small indoor networks with stable Wi-Fi backhaul such as offices, retail stores and hospitality environments.
Setu SCX2	12–24 V (Power)				Designed for mission-critical deployments where Ethernet, Wi-Fi and LTE failover ensure uninterrupted connectivity in industrial and multi-site facilities.

What's in the Box?

Make sure box contains following items before installations



Setu



**Mounting
Clip***



**User manual &
Warranty Card**



**IP
Extender***



**Mounting
Screws**

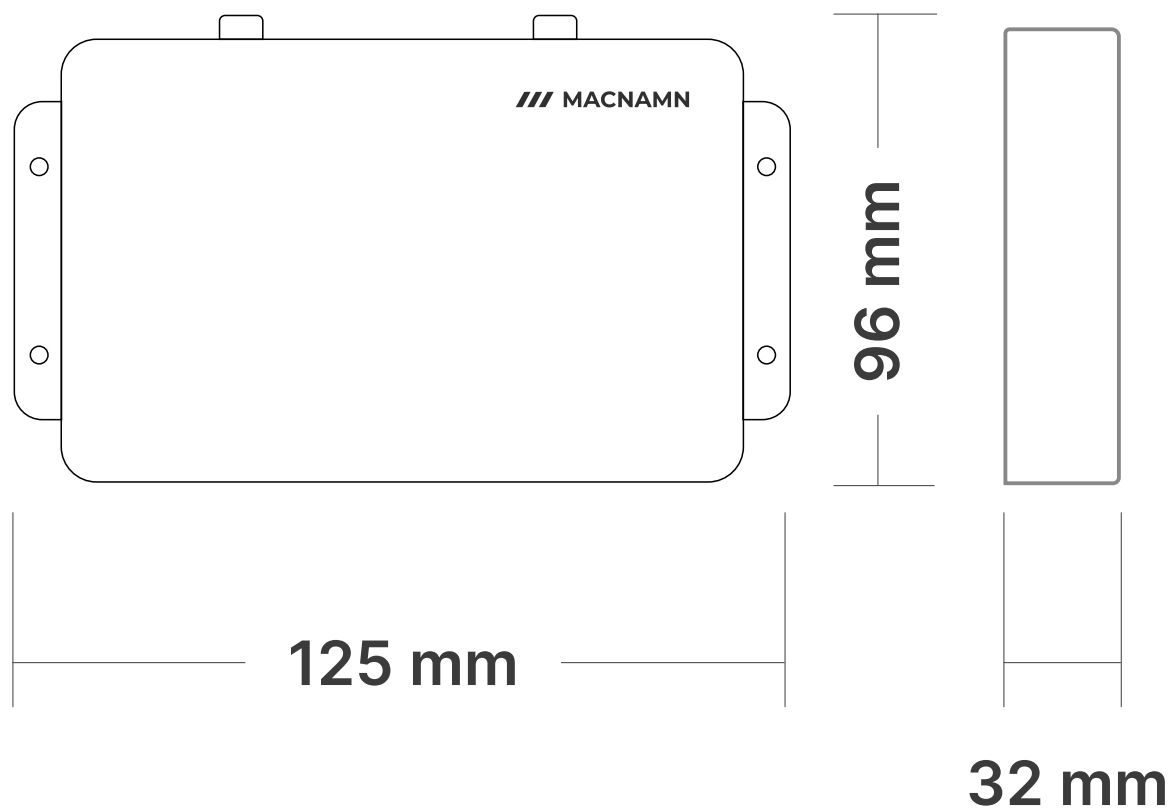


**Power
Adapters***

i Note

Accessories must be purchased separately & may not be included in the box.

Setu SCX Dimensions



Frequently Asked Questions

What is the Setu SCX used for?

Setu SCX is a compact single-channel indoor LoRa gateway offering a low-cost access point for small IoT deployments in commercial and enterprise spaces. Typical applications include Smart Buildings & Offices, Hospitals & Healthcare Facilities, Retail & Shopping Centers, Hotels & Hospitality Spaces, Smart Campuses & Education and Industry 4.0 & Factory Floors.

Which LoRa channels and frequency bands does the Setu SCX support?

The Setu SCX is a single-channel gateway: Single-Channel (1 × Uplink / 1 × Downlink). Supported frequency band: IN865. It works with Macnman LoRa Nodes & Controllers. Receive sensitivity: Down to –136 dBm @ SF12, BW 125 kHz.

Which backhaul options does the Setu SCX have?

The Setu SCX offers Wi-Fi and cellular LTE (optional) backhaul. Cellular bands: B1 / B3 / B5 / B7 / B8 / B20 / B28.

How is the Setu SCX powered?

The Setu SCX is powered from 9–36 V PoE.

What is the IP rating and operating temperature of the Setu SCX?

The Setu SCX has an ingress protection rating of IP35 and an operating temperature range of –40°C to +80°C. Operating humidity: 0% to 100% RH (non-condensing). Mounting options: Pole Mounting / Wall Mounting.

Related Products and Resources

- Related datasheets: [IndiLoRa X single-channel indoor LoRa gateway](#), [MacLink Gen-2 industrial outdoor LoRaWAN gateway](#), [Setu LX Series indoor LoRaWAN gateway](#), [FireGaurd Gen 1 fire equipment health monitoring controller](#)
- Help: [Warranty terms and conditions](#) and [contact Macnman support](#)